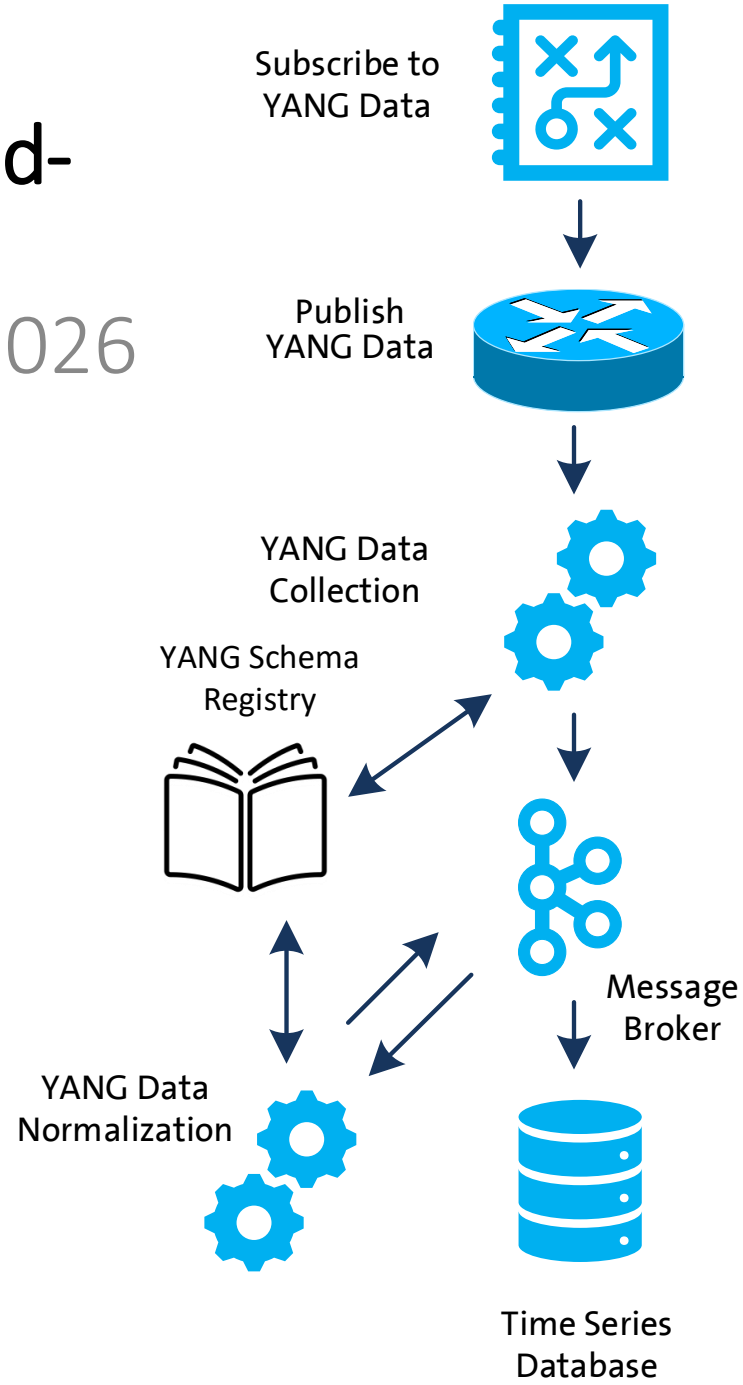
Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 127 Hackathon, November 14/15th 2026





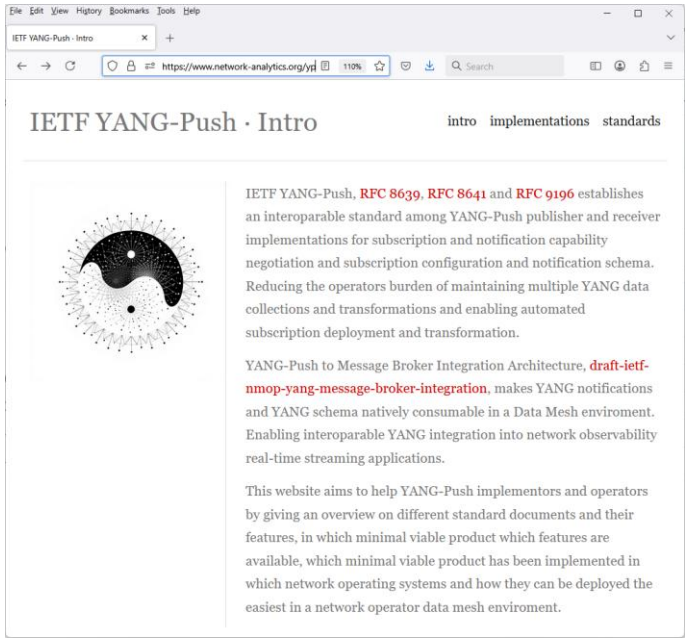
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 6 YANG-Push Publishers
- 3 YANG-Push Receivers
- 3 YANG-Push Network Telemetry Message
- 2 YANG Message Broker Producer
- 1 YANG Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

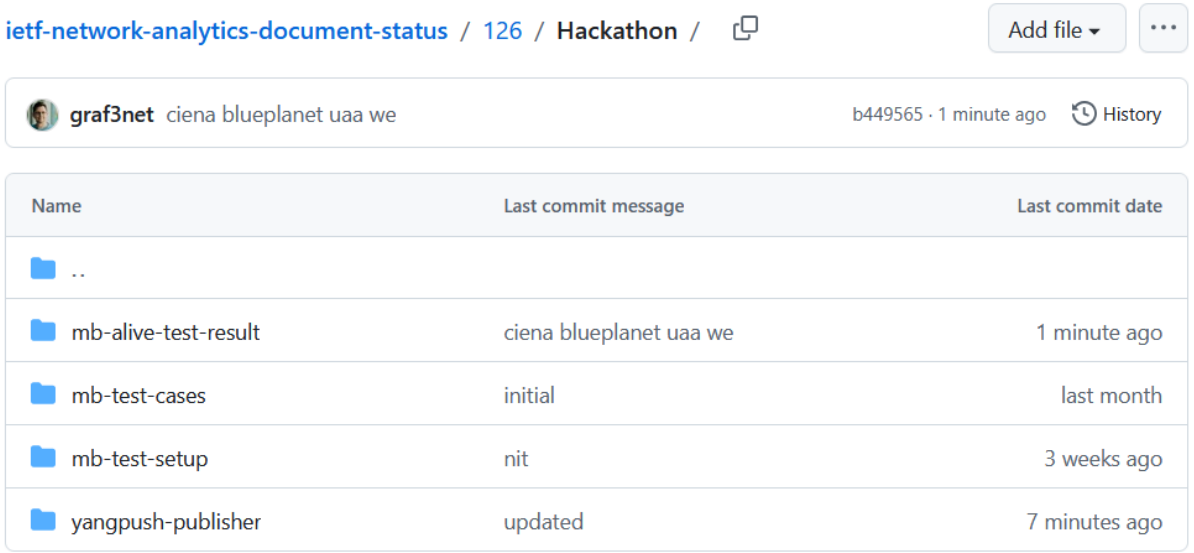
https://www.network-analytics.org/yp/how-to-deploy.html

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router),MA (OLT), Arrcus Arcos and Ductus DMAP
- YANG-Push Receiver – NetCalyx, NetGauze® and Pmacct
- YANG Message Broker Producer – NetCalyx, NetGauze®
- YANG Message Broker Consumer – NetCalyx, NetGauze®, Ciena Blue Planet UAA, Cisco Crosswork
- YANG Schema Registry – Apache Kafka YANG Plugin
- Libyang - https://github.com/CESNET/libyang/releases/tag/v5.4.9
- Yangkit - https://github.com/yang-central/yangkit/
- Wireshark - udp-notif dissector

Hackathon – Repositories

The results of the end-to-end tests

- Netconf RPCs and YANG-Push JSON and CBOR encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation

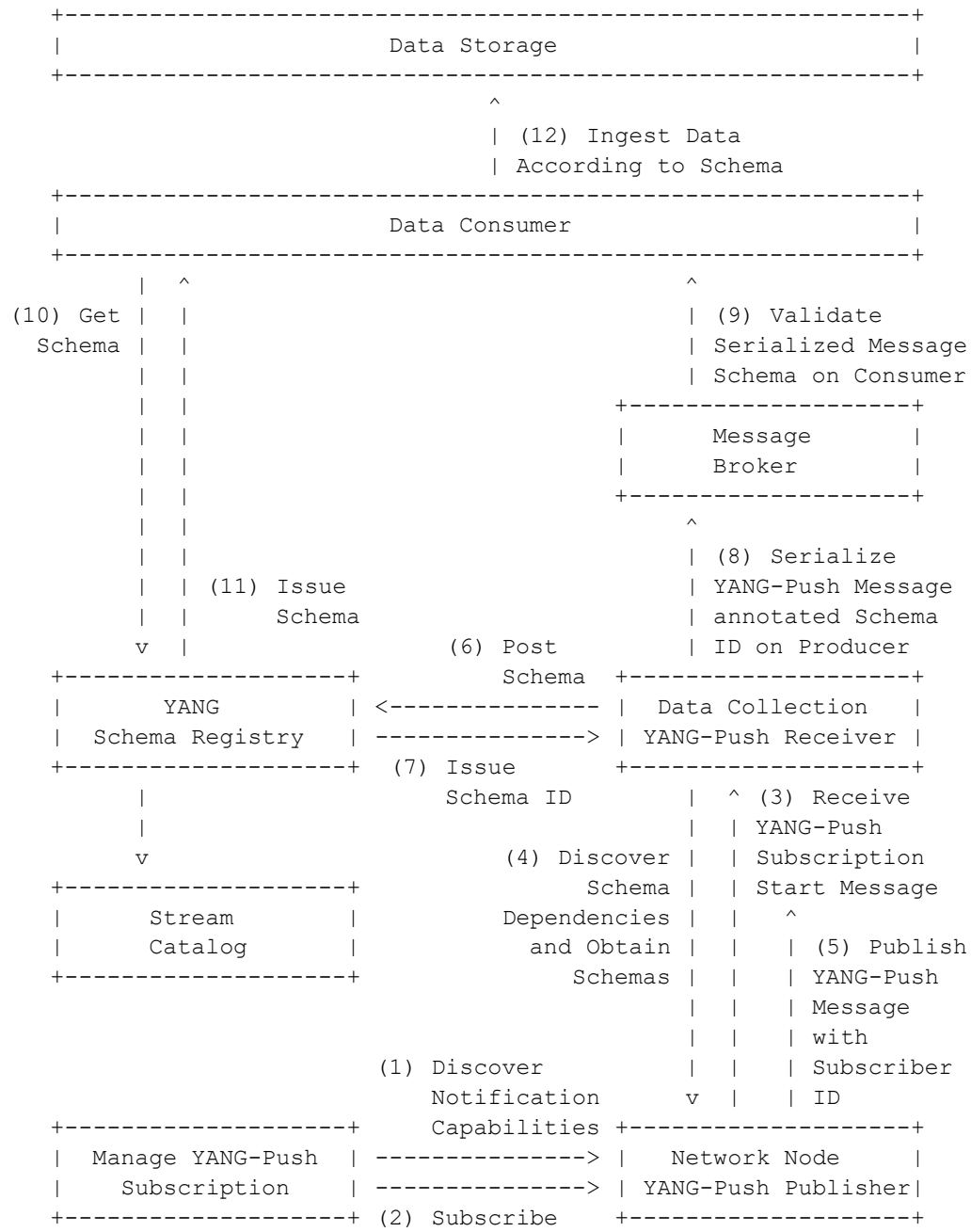


https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/127/Hackathon/

An Architecture for YANG-Push to Message Broker Integration

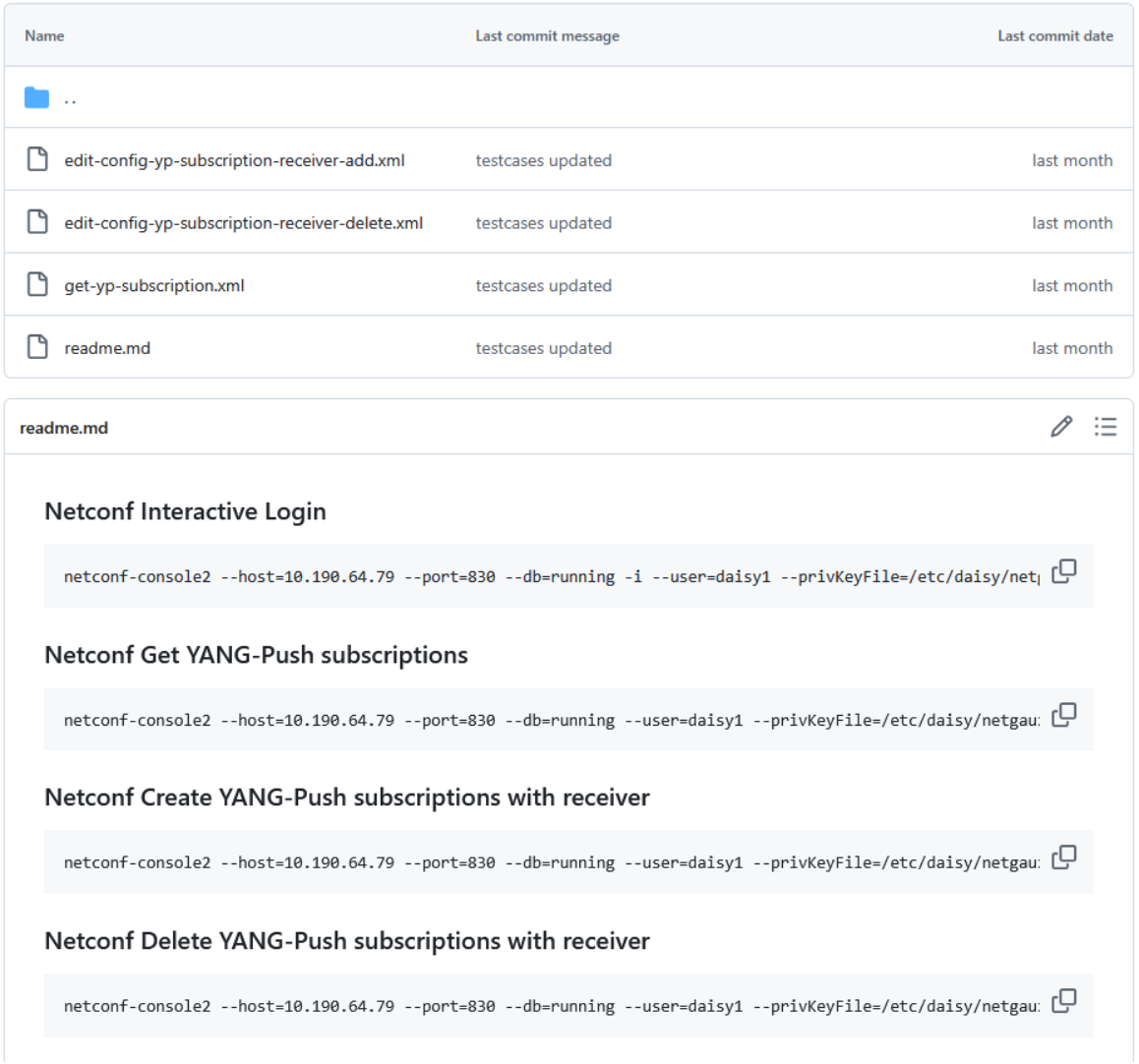
draft-ietf-nmop-yang-message-broker-integration
draft-ietf-nmop-message-broker-telemetry-message
draft-ietf-nmop-yang-message-broker-message-key
draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
  RFC 8639
- Subscription to YANG Notifications for Datastore Updates
  RFC 8641
- UDP-based Transport for Configured Subscriptions
  draft-ietf-netconf-udp-notif
- Subscription to Distributed Notifications
  draft-ietf-netconf-distributed-notif
- Extensible YANG Model for YANG-Push Notifications
  draft-ietf-netconf-notif-envelope
- Support of Versioning in YANG Notifications Subscription
  draft-ietf-netconf-yang-notifications-versioning
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
  RFC 9196
- YANG Notification Transport Capabilities
  draft-ietf-netconf-yp-transport-capabilities
- YANG Library
  RFC 8525
- Augmented-by Addition into the IETF-YANG-Library
  RFC 10035
- Validating anydata in YANG Library context
  draft-netana-netmod-yang-anydata-validation
- Encoding of Data Modeled with YANG in the CBOR
  RFC 9254



YANG-Push Publisher - Subscription

- Netconf RPC calls to create, delete and obtain YANG-Push subscriptions and receiver configurations.



Name	Last commit message	Last commit date
..		
edit-config-yp-subscription-receiver-add.xml	testcases updated	last month
edit-config-yp-subscription-receiver-delete.xml	testcases updated	last month
get-yp-subscription.xml	testcases updated	last month
readme.md	testcases updated	last month

readme.md

Netconf Interactive Login

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running -i --user=daisy1 --privKeyFile=/etc/daisy/net
```

Netconf Get YANG-Push subscriptions

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Create YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Delete YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

YANG-Push Publisher - Notifications

- Subscription state change notifications as JSON and PCAP.
- Push update and change-update notifications as JSON and PCAP.

Name	Last commit message	Last commit date
..		
1-push-update.json	testcases updated	last month
1-subscription-started.json	testcases updated	last month
1-yp-subscripton.pcap	testcases updated	last month
2-push-update.json	testcases updated	last month
2-subscription-started.json	testcases updated	last month
2-yp-subscripton.pcap	testcases updated	last month
27-push-update.json	testcases updated	last month
27-subscription-started.json	testcases updated	last month
27-yp-subscripton.pcap	testcases updated	last month
readme.md	testcases updated	last month

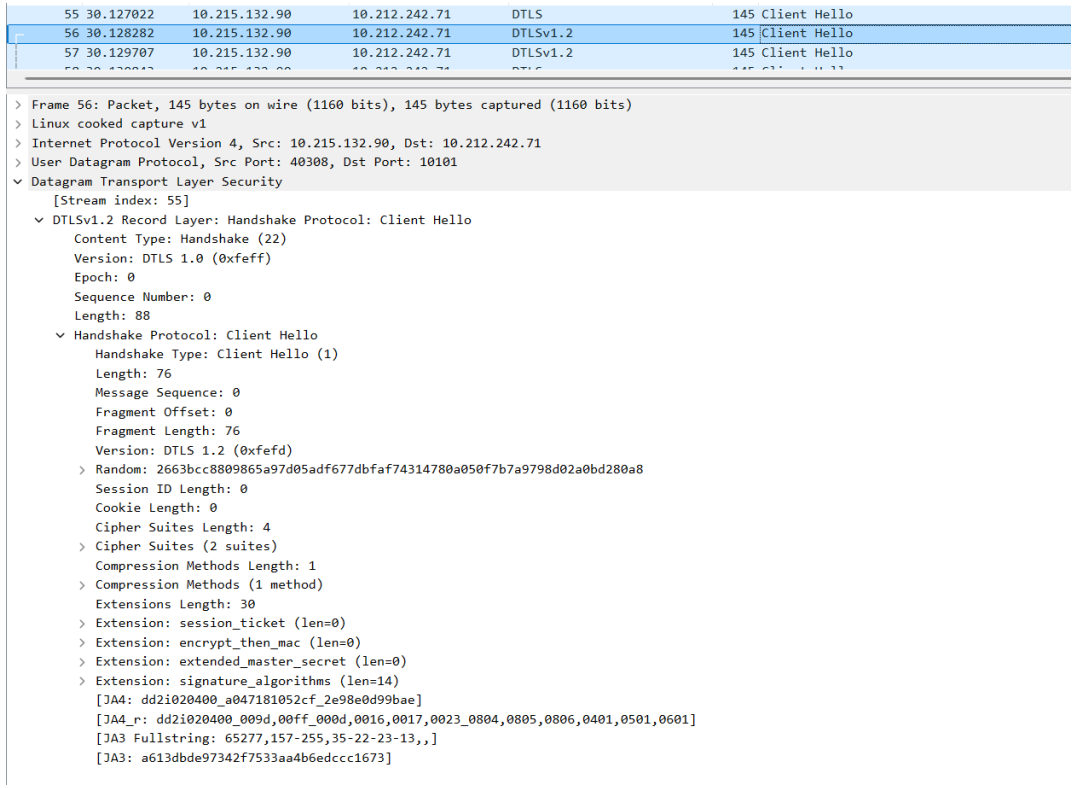
readme.md

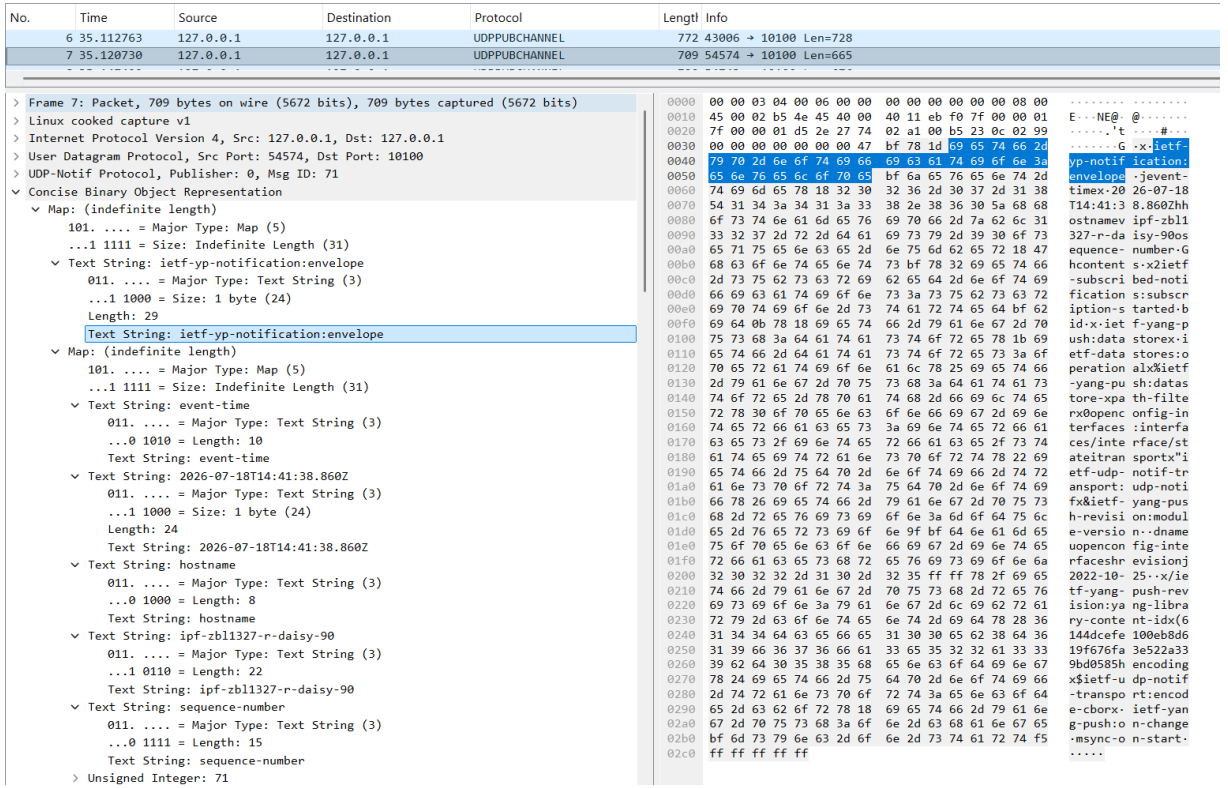
YANG-Push Notifications

subscription-started, push-update and push-change-update notifications in raw and json sorted by subscription id. pcap requires https://github.com/network-analytics/udp-notif-dissector Wireshark dissector.

YANG-Push UDP-Notif – DTLS and CBOR

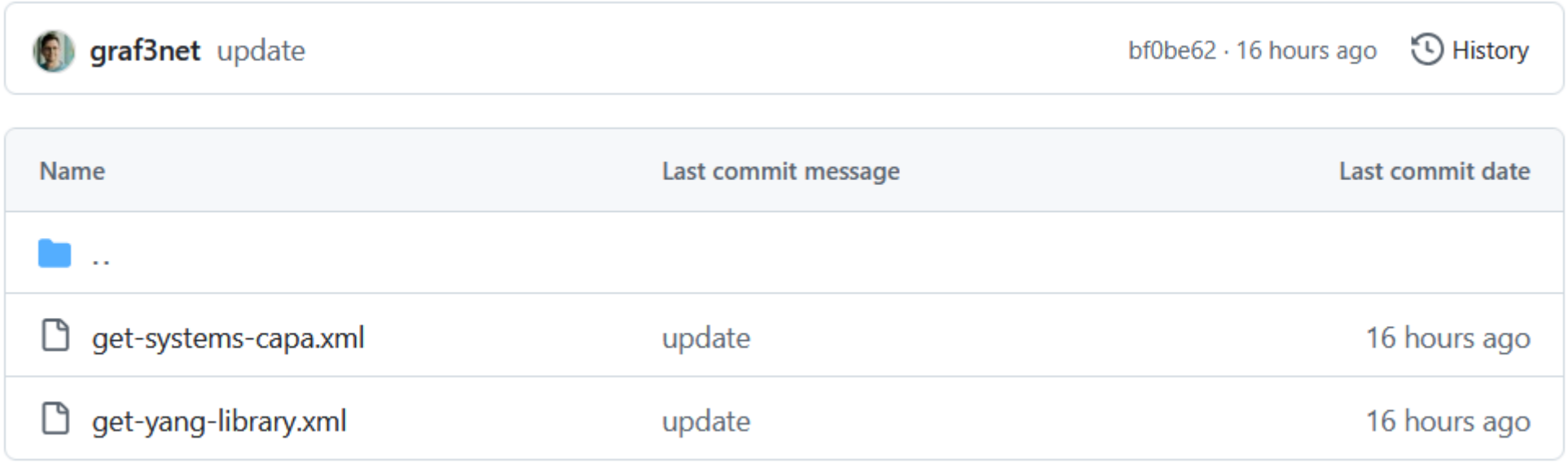
- With Cisco IOS XR, 6Wind VSR and Arrcus Arcos we have now three implementations support CBOR Name Identifiers
- With Cisco IOS XR we have the first implementation support DTLS 1.2 in UDP-Notif.





YANG-Push Capabilities & YANG Library - RPC

- Netconf XML encoded reply for RFC 9196 YANG-Push capabilities.
- Netconf XML encoded reply for RFC 8525 with RFC 10035.



graf3net update bf0be62 · 16 hours ago History

Name	Last commit message	Last commit date
..		
get-systems-capa.xml	update	16 hours ago
get-yang-library.xml	update	16 hours ago

YANG-Push

IETF 127 Publisher Implementation Status



	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open-Source	Ductus DMAP
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓		✓
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif with DTLS				✓			
draft-ietf-netconf-distributed-notif	✓	✓	✓		na		
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓		P
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓	✓		
RFC 8525 YANG Library	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓	✓
RFC 9196 System and Notification Capabilities			✓	P	✓		✓
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓		
RFC 9254 CBOR Named Identifiers	✓			✓	✓		

Green marked describes new capability at IETF 127. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 127 Module Implementation Status



	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	✓	✓		
ietf-hardware.yang (inventory, state and statistic)	P	✓	✓		
ietf-alarms.yang (state)	✓	✓	✓		
ieee802-dot1ab-lldp.yang (inventory and state)	P	✓			
ieee802-dot1ax.yang (inventory, state and statistic)		✓			
ietf-bfd-ip-sh.yang (state)		P			
ietf-bfd-ip-mh.yang (state)		P			
ietf-bfd-lag.yang (state)		P			
ietf-isis.yang (state)		P			

Green marked describes new capability at IETF 127. "P" to partially implemented.

YANG Message Broker

IETF 127 Implementation Status



	NetCalyx	NetGauze®	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓	✓			
YANG-Push Receiver	✓	✓	✓			
YANG Schema Retrieval	✓	✓				
YANG Message Broker Producer	✓	✓				
YANG Schema Registry	na	na	na	✓	na	na
YANG Message Broker Consumer	✓	✓			✓	✓
Validate Telemetry Message against YANG Schema	✓	✓			✓	P
draft-netana-netmod-yang-anydata-validation	P	P			P	
YANG Schema Comparison	na	na	na	✓	na	na

Green marked describes new capability at IETF 127. "P" to partially implemented.

Next Steps from IETF 126 Hackathon…

- Support building RFC 8525 YANG Library in Java Message Broker Consumer
- Support draft-netana-netmod-yang-anydata-validation and RFC 8525 YANG library in yangkit
- draft-netana-netmod-yang-anydata-validation schema anchor proof of concept in libyang



Next Steps for IETF 128 Hackathon…

- draft-netana-netmod-yang-anydata-validation schema anchor support in NetCalyx
- draft-netana-netmod-yang-anydata-validation support in yang-tools
- YANG identityref schema dependency resolution
- Support YANG-Push Message Broker Topic Naming with dynamic Apache Kafka topic creation API

Thanks to…

- Jiaxin PAN – Huawei
- Zhengliang Li – Huawei
- Xiao Chen – Huawei
- Qin Wu - Huawei
- Rob Wilton – Cisco
- Dan Voyer – Cisco
- Scott Huang - Cisco
- Muthumayan Madhayyan - Cisco
- Jérémie Leska – 6Wind
- Per Anderson - Ductus
- Deepya Mandadi – Ciena
- IJsbrand Wijnands – Arrcus
- Irfan Mohammad – Arrcus
- Benoit Claise - Everything OPS
- Paolo Lucente – Pmacct
- Vivekananda Boudia – INSA Lyon
- Alex Huang Feng – T-Systems
- Mohamed Kheir - SingleStore
- Helmar Becker - ClickHouse
- Maxence Younsi – Swisscom
- Leonardo Rodoni - Swisscom
- Ahmed Elhassany – Swisscom
- Thomas Graf – Swisscom

Want to know more then join us at…

- **Side meeting** on **Monday 11:00 – 12:00** at Plaza A for end-to-end demo on YANG-Push to Message Broker and time series database integration.
- **NMOP** on **Thursday 16:30 – 18:30** at Golden Gate 3 for Message Broker, BMP Telemetry Message, YANG anydata validation document updates and hackathon.
- **NETMOD** on **Monday 16:30 – 18:30** at Golden Gate 4-5 for YANG anydata validation.